surprisingly, Linde had called this approach *chaotic inflation.* Many physicists consider it the most convincing realization of the inflationary paradigm.

12. Those familiar with the history of this subject will realize that the excitement over Guth's discovery was generated by its solutions to key cosmological problems, such as the horizon and flatness problems, as we describe shortly.

13. You might wonder whether the electroweak Higgs field, or the grand unified Higgs field, can do double duty—playing the role we described in Chapter 9, while also driving inflationary expansion at earlier times, before forming a Higgs ocean. Models of this sort have been proposed, but they typically suffer from technical problems. The most convincing realizations of inflationary expansion invoke a new Higgs field to play the role of the inflaton.

14. See note 11, this chapter.

15. For example, you can think of our horizon as a giant, imaginary sphere, with us at its center, that separates those things with which we could have communicated (the things within the sphere) from those things with which we couldn't have communicated (those things beyond the sphere), in the time since the bang. Today, the radius of our "horizon sphere" is roughly 14 billion light-years; early on in the history of the universe, its radius was much less, since there had been less time for light to travel. See also note 10 from Chapter 8.

16. While this is the essence of how inflationary cosmology solves the horizon problem, to avoid confusion let me highlight a key element of the solution. If one night you and a friend are standing on a large field happily exchanging light signals by turning flashlights on and off, notice that no matter how fast you then turn and run from each other, you will *always* be able subsequently to exchange light signals. Why? Well, to avoid receiving the light your friend shines your way, or for your friend to avoid receiving the light you send her way, you'd need to run from each other at faster than light speed, and that's impossible. So, how is it possible for regions of space that were able to exchange light signals early on in the universe's history (and hence come to the same temperature, for example) to now find themselves beyond each other's communicative range? As the flashlight example makes clear, it must be